

JAVA LAB WEEK 3

EXPERIMENT : PRIME PALINDROME

1. Basic Information

Experiment No.: 01

Date: 17/09/2026

Title: Program to Check Whether a Given Number is Prime Palindrome or Not

2. Objective / Aim

To write and execute a Java program to check whether the given number is both **prime** and **palindrome** using loops and conditional statements.

3. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Hard Disk: Minimum 10 GB free space
- Keyboard and Monitor

Software Requirements

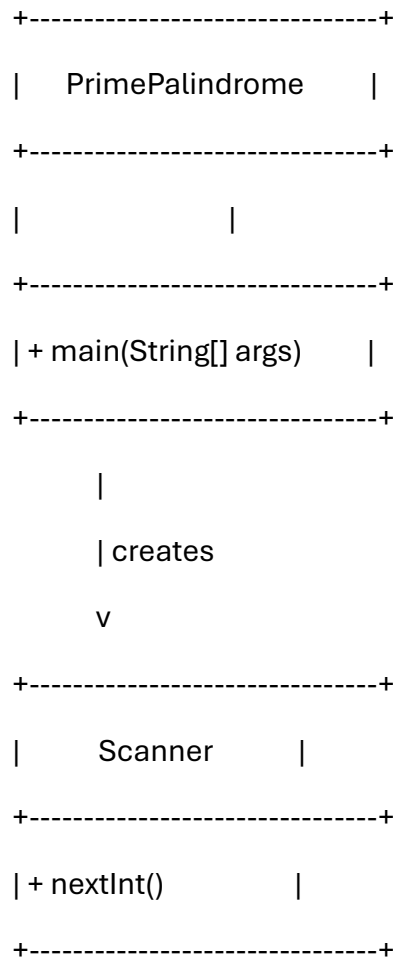
- Operating System: Windows/Linux/macOS
- JDK: JDK 8 or above
- IDE: IntelliJ IDEA, Eclipse, NetBeans, or Visual Studio Code
- Command Prompt/Terminal

4. Algorithm / Logic Description

1. Start the program.
2. Import the Scanner class.
3. Create a Scanner object to read the number from the user.
4. Read the number n.
5. Initialize count to 0.
6. Use a for loop from 1 to n to find the number of divisors of n.
7. Increment count whenever n is exactly divisible by i.
8. Initialize rev to 0 and store the original number in temp.

9. Use a while loop to reverse the digits of the number.
10. Check whether count is equal to 2 and rev is equal to n.
11. If both conditions are true, display Prime Palindrome.
12. Otherwise, display Not Prime Palindrome.
13. Stop the program.

5. UML Class Diagram



6. Source Code

```

import java.util.Scanner;

class PrimePalindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
    
```

```

System.out.print("Enter a number: ");

int n = sc.nextInt();

int count = 0, rev = 0, temp = n;

for (int i = 1; i <= n; i++) {
    if (n % i == 0)
        count++;
}

while (temp > 0) {
    rev = rev * 10 + temp % 10;
    temp = temp / 10;
}

if (count == 2 && rev == n)
    System.out.println("Prime Palindrome");
else
    System.out.println("Not Prime Palindrome");
}
}

```

7. Compilation & Execution Commands

Java Filename: PrimePalindrome.java

Compilation

```
javac PrimePalindrome.java
```

Execution

```
java PrimePalindrome
```

8. Output Screen

Sample Output

Enter a number: 131

Prime Palindrome

Another Sample Output

Enter a number: 123

Not Prime Palindrome

9. Result

The program was successfully implemented and executed. The experiment demonstrates the use of **Scanner, for loop, while loop, variables, arithmetic operators, and conditional statements** to check whether a given number is a prime palindrome.

EXPERIMENT : STUDENT MARKS & AVERAGE

1. Basic Information

Experiment No.: 02

Date: 17/09/2026

Title: Program to Find Total Marks and Average Using Command-Line Arguments

2. Objective / Aim

To write and execute a Java program to read the marks of five subjects using **command-line arguments** and calculate the **total marks and average marks**.

3. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Hard Disk: Minimum 10 GB free space
- Keyboard and Monitor

Software Requirements

- Operating System: Windows/Linux/macOS
- JDK: JDK 8 or above
- IDE: IntelliJ IDEA, Eclipse, NetBeans, or Visual Studio Code
- Command Prompt/Terminal

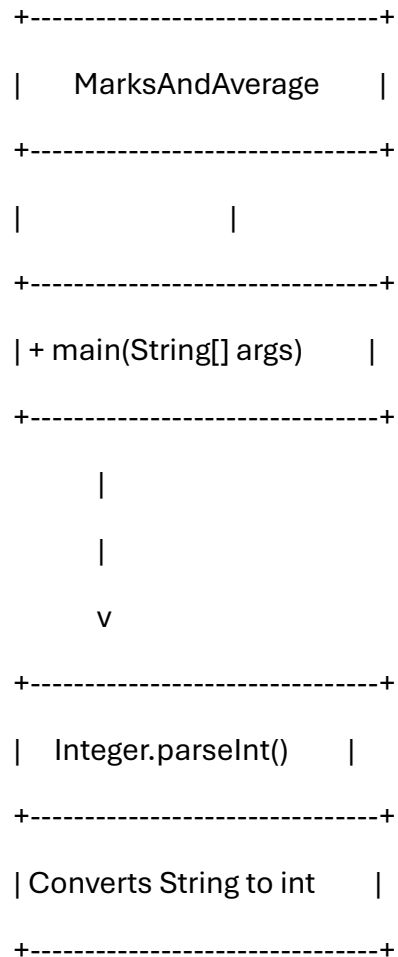
4. Algorithm / Logic Description

1. Start the program.
2. Create the class MarksAndAverage.
3. Define the main() method.
4. Declare five integer variables s1, s2, s3, s4, and s5.
5. Read the five subject marks from the command-line arguments.
6. Convert each command-line argument from String to int using Integer.parseInt().
7. Add the five subject marks to calculate the total.
8. Divide the total by 5 to calculate the average.
9. Display the total marks.

10. Display the average marks.

11. Stop the program.

5. UML Class Diagram



6. Source Code

```
public class MarksAndAverage {
    public static void main(String[] args) {
        int s1,s2,s3,s4,s5;
        s1=Integer.parseInt(args[0]);
        s2=Integer.parseInt(args[1]);
        s3=Integer.parseInt(args[2]);
        s4=Integer.parseInt(args[3]);
        s5=Integer.parseInt(args[4]);
```

```
        System.out.println("the total marks is:"+(s1+s2+s3+s4+s5));

        System.out.println("the average of he total marks is:"+((s1+s2+s3+s4+s5)/5));

    }

}
```

7. Compilation & Execution Commands

Java Filename: MarksAndAverage.java

Compilation

```
javac MarksAndAverage.java
```

Execution

For example, to give marks 80, 75, 90, 85, 70:

```
java MarksAndAverage 80 75 90 85 70
```

8. Output Screen

the total marks is:400

the average of he total marks is:80

Calculation

Total = 80 + 75 + 90 + 85 + 70

= 400

Average = 400 / 5

= 80

9. Result

The program was successfully implemented and executed. The experiment demonstrates the use of **command-line arguments, Integer.parseInt(), variables, arithmetic operators, and output statements** to calculate the total and average marks of five subjects.

OVERALL RESULT

The given Java programs were successfully implemented and executed.

The experiments demonstrate the following Java concepts:

- Scanner for user input

- Command-line arguments
- `Integer.parseInt()`
- for loop
- while loop
- Variables and data types
- Arithmetic operators
- Conditional statements
- `main()` method
- Console output